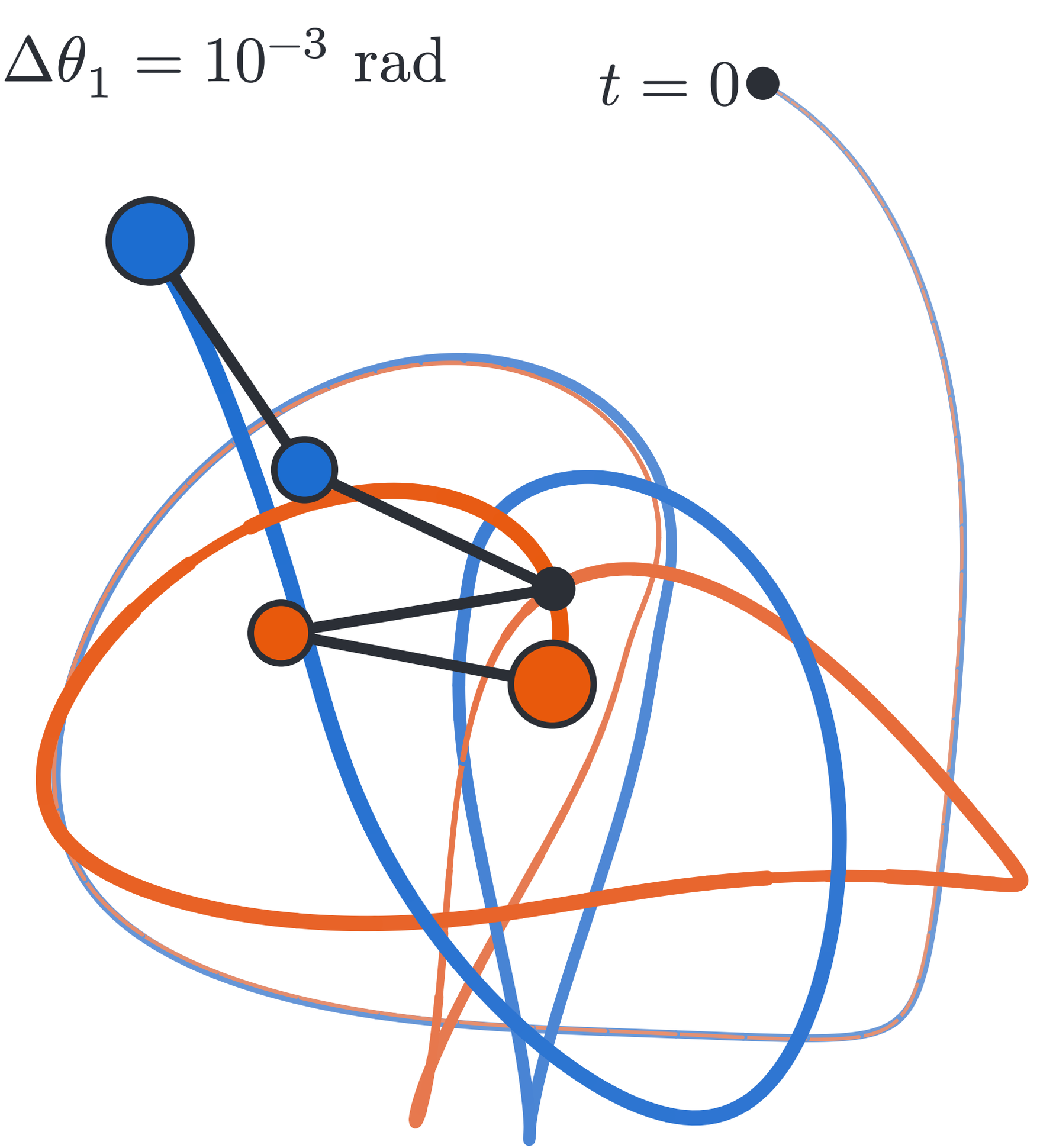
$\Delta\theta_1 = 10^{-3}$ rad
$t = 0$